CDFD Runtime Paper 07: Autonomous Discovery, Hypothesis Triage, and Falsification

Steve Bico Mujjabi, MD
Independent Researcher
Founder, Vura Labs
Kampala, Uganda
ORCID: 0009-0001-0556-5516

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Abstract

Executive synthesis. The discovery layer is a hypothesis factory, not an oracle. It perturbs runtime states, extracts features, proposes candidate regimes, and records falsification targets. This paper aligns the discovery manuscripts with the actual local experiment surface in `cdfd_runtime/experiments`: OOL phase mapping, finance-climate coupling, schema invention, vacuum hysteresis, knot stability, adaptive-surface instability, frontier sweeps, and gigamarathon coverage.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, Ω denotes a domain or state space, and Λ denotes the Life Number where used. The coefficients α, β, and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or redistribution. The symbols δ and ϵ denote perturbation or tolerance scales, while η, λ, μ, ν, ρ, σ, τ, and ϕ denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Discovery Discipline

Autonomous discovery is useful only when it remains falsifiable. The CDFD The Runtime discovery layer does not turn a pattern into a law merely because the engine generated it. A discovery candidate needs:

- a script or CDFL model;
- saved output;
- parameter values;
- a pattern label;
- a falsification condition;
- a rerun command.

3 Discovery Modules

The discovery package contains feature extraction, hypothesis, generator, knowledge, sensitivity, detector, validator, and causal-discovery modules. The workflow is:

1. generate or perturb a state;
2. run bounded dynamics;
3. extract features such as mean Ψ_s, variance, collapse count, and stability index;
4. assign a candidate pattern label;
5. write a report or output file;
6. attempt falsification by changing parameters or null-coupling.

4 Selected Local Discoveries

4.1 Discovery A: OOL phase boundary

`experiments/notebooks/discovery_ool_phase.py` sweeps a two-dimensional prebiotic parameter space. Its selected HDF5 output contains grid points near $\Lambda = 1$. This is useful as a cross-Part-II phase-boundary bridge and as a test target for the Life Number. It is not empirical proof of abiogenesis.

4.2 Discovery B: finance-climate coupling

`experiments/notebooks/discovery_finance_climate.py` models a coupled finance-climate toy system and then runs causal discovery over the time series. The report frames the edge from financial Ψ volatility to climate constraint as a candidate causal result. The necessary null test is explicit: set coupling to zero and verify that the edge disappears.

4.3 Discovery C: schema invention

The schema-invention demonstration tests whether the knowledge layer can assign a new candidate label to a high-Ψ_s anomaly. The narrow result is software capability: novelty detection and candidate schema recording. It is not, by itself, proof of a new physical regime.

4.4 Discovery D: vacuum hysteresis

`experiments/notebooks/discovery_vacuum_hysteresis.py` tests whether a high-flux pulse leaves localized M_s memory. The selected output records a center M_s peak of 15.5843. The proper status is candidate simulation result and measurement-protocol proposal.

4.5 Discovery E: n=7 knot stability

`experiments/notebooks/discovery_knot_symmetry.py` embeds a T(2,7) constraint pattern and tracks a stability index toward the CODATA 2022 $\chi^* = 137.035999177$ target. The release-selection note says to use it as a weak higher-knot self-stabilization stress test, not a proof.

4.6 Discovery F: adaptive-surface instability

`experiments/notebooks/discovery_adaptive_surface.py` produces an overloaded run. Because Ψ_s grows from approximately 5.15 to approximately 979 in the selected output, this is presented as an instability boundary and a safety test, not as an application success.

5 Causal Graph from Run History

`engine/causal_graph.py` builds Granger-style directed edges from simulation time series without an LLM. The CLI and optional web studio can present these edges as triage hints: which series preceded which under the current model. Optional LLM provider calls may later summarize a saved result, but they do not create the edge, alter the finite audit, or turn a candidate edge into validation. A persistent edge after decoupling controls remains an artifact, not a discovery.

6 Frontier and Gigamarathon Sweeps

`experiments/outputs/frontier_sweep.h5` summarizes 2,500 parameter points and 2,487 non-trivial candidate records. This is hypothesis triage: which regions deserve closer study. `gigamarathon_1000_results.h5` covers 10 domains by 100 points each. This is coverage evidence for the discovery engine, not physical validation.

7 Promotion Ladder

A candidate discovery moves through this ladder:

1. simulation anomaly;
2. reproducible runtime diagnostic;
3. parameter-sensitivity report;
4. falsification attempt;
5. independent dataset or physical experiment;
6. domain-specific validation.

8 Conclusion

The discovery layer is valuable because it can search wider than a human can manually search. Its danger is overstatement. The runtime papers present discoveries as candidate simulation results with falsification handles until external validation exists.

References

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